Muhammad Rafi Arsya

Software Engineering Student

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Profile

Software Engineering student at Universiti Malaya who takes ideas all the way to live products. Built and runs two student platforms, CampusBay and UMOVE, end to end: Next.js and React front ends, Node and PostgreSQL back ends, Cloudflare Workers, and a self-hosted Linux server behind a Cloudflare Tunnel. Also works in applied AI/ML (RAG, CNNs written from scratch). Seeking a Data Engineering or Software Engineering internship.

Education

Universiti Malaya (UM), Malaysia *Since 2025*
B.Sc. Computer Science (Software Engineering)

- Coursework: Data Structures & Algorithms, Machine Learning, Network Technology, Human Computer Interaction, Computer Systems Architecture.

International Program ICBS, Riau, Indonesia *2022 to 2025*
Senior High School

- International curriculum with English-medium instruction.
- Active in student leadership and language programs.

Experience

PEKOM: Computer Science Society, Universiti Malaya *Since 2026*
Technical Executive Committee

- Serving in the Website Division, building and maintaining the society's web platforms.
- Supporting faculty-wide events across the Faculty of Computer Science & IT.

PPI Malaysia (Indonesian Students Association) *Since 2025*
Head of Website Division, Data & Information System Bureau

- Full ownership of the official PPI Malaysia website: development, maintenance, and continuous improvement.
- Serving Indonesian students across Malaysia.

PPI University of Malaya *Since 2026*
Head of Department, Art Exhibition at IDFEST

- Directing the Art Exhibition division for IDFEST, an international event showcasing Indonesian heritage at Universiti Malaya, from concept and curation through to delivery.
- Leading the department end to end: theme and curation, artist and submission coordination, exhibition layout, and delivery on event day.

PPI University of Malaya *Since Jun 2026*
Website Development Committee at IDFEST

- Building the official IDFEST website with Next.js 16, React 19 and Tailwind CSS v4, translating Figma designs into production-ready pages.
- Owning the homepage sections and animated timeline, hero responsiveness across breakpoints, and shared navigation and mascot components reused by the Art Exhibition and Musical Theater pages.

Volunteering

Universiti Malaya Week of Welcome (UM-WOW) *2026*
Buddy, Student Affairs Department

- Welcomed and guided new students through orientation week, registration and their first days on campus at Universiti Malaya.

Faculty of Computer Science & Information Technology, Universiti Malaya *1 to 5 September 2026*
Student Buddy, Inbound Mobility Programme 2026

- One of a four-person committee running a five-day FSKTM programme hosting students from Chulalongkorn University, Thailand's top-ranked university, in Kuala Lumpur.
- Served as a Student Buddy to the visiting delegation and was responsible for documentation across the whole programme, covering sessions and daily activities on all five days.

LARAS, PPI Malaysia *2026*
Transportation Team & Field Committee

- Handled logistics, on-ground coordination and event-day operations for LARAS, PPI Malaysia's annual gathering.

Selected Projects

CampusBay: Student Marketplace *2026*
Live marketplace where students list items or post requests and close deals on WhatsApp. Next.js 16 on Cloudflare Workers with D1 and Drizzle, instant updates through a Durable Object, EN/MS/ID search and a full moderation panel. Built and run solo.

UMOVE: Campus Errand Delivery *2026*
Students post errands and verified student runners deliver them, paid on delivery. React 19 front end on Cloudflare Workers; Node, Hono and PostgreSQL API in Docker on a mini PC behind Cloudflare Tunnel, with a live board over Server-Sent Events.

PaperMind: Local RAG Paper Analyzer *2026*
Ask questions of any PDF with answers grounded in the text. FastAPI, ChromaDB and Ollama, fully offline, in Docker Compose on a self-hosted Linux server.

AI Digit Recognizer: CNN From Scratch *2026*
Convolutional network written entirely in NumPy, from convolutions to backpropagation and Adam. 99.1% on the MNIST test set, served by Flask with a layer-by-layer visualizer.

Nalar: Blog with a Custom CMS *2026*
Node.js, Express and PostgreSQL blog with JWT auth, Markdown editing and scheduled publishing. Self-hosted behind Cloudflare Tunnel, managed with PM2.

snip: Serverless URL Shortener *2026*
Click-tracking URL shortener on API Gateway, Lambda and DynamoDB, defined as infrastructure as code with AWS SAM and least-privilege IAM per function.

Crop Disease Detector *2026*
Classifies 38 plant diseases from a leaf photo. MobileNetV2 transfer learning with two-phase fine-tuning in TensorFlow/Keras, about 97% accuracy, deployed on Hugging Face Spaces.

IDFEST 2026 Website *2026*
Official site for PPI UM's cultural festival, built as a team in Next.js 16 and Tailwind v4. Owned the homepage timeline, responsive hero and shared navigation.

Skills

Core: TypeScript, JavaScript, Python, React, Next.js, Node.js, PostgreSQL / SQL, Docker, Cloudflare, Git

Also used in projects:

Languages: Java, PHP, Bash

Frontend: Tailwind CSS, Vite, Bootstrap 5, Chart.js / Recharts, PWA, Chrome Extensions (Manifest V3)

Backend: Express, Hono, FastAPI, Flask, WebSockets / Socket.IO, Server-Sent Events, Cloudflare Workers & Durable Objects, Zod, JWT & Google OAuth

Data: SQLite / Cloudflare D1, Drizzle ORM, Supabase, DynamoDB, ChromaDB

Infrastructure & Testing: Docker Compose, Cloudflare Tunnel, AWS (Lambda, API Gateway, SAM), NGINX, Ubuntu self-hosting, GitHub Actions CI, Playwright

AI / ML: NumPy, TensorFlow / Keras, OpenCV, MediaPipe, scikit-learn, pandas, RAG with Ollama, Hugging Face, Claude API

Certifications & Courses

- Ultimate AWS Certified Developer Associate (DVA-C02), Udemy course completion, Stephane Maarek, June 2026
- Ultimate AWS Certified AI Practitioner (AIF-C01), Udemy course completion, Stephane Maarek, May 2026
- Foundations: Data, Data, Everywhere, Google Data Analytics Certificate, Coursera, September 2026
- IELTS Academic, Band 5.5 (CEFR B2), British Council IDP, June 2025
- Additional coursework, around 570 hours across 2025 and 2026: freeCodeCamp JavaScript Algorithms & Data Structures (300 hours), The Complete JavaScript Course (Jonas Schmedtmann), full-stack bootcamps with Angela Yu and Cloudaffle, Advanced SQL (Maven Analytics), The Complete SQL Bootcamp (Jose Portilla), Spring AI Fast Track, and Cloud & Gen AI on AWS plus Spec-Driven Development with Kiro (Dicoding)

Languages

Bahasa Indonesia: Native | English: Professional working proficiency | Malay: Professional working proficiency